Hints: Methods of proof – 6. Contradiction and proofs of impossibility

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Suppose $2^a, 2^b, 2^c$ with $a < b < c$ form an arithmetic progression: $2^a + 2^c = 2^{b+1}$. Divide by 2^a and look at parities.

Hint 2. $1 + 2^{c-a}$ is odd and greater than 1, while 2^{b-a+1} is even. Contradiction (this is a 2-adic valuation argument).

Problem 2

Hint 1. Pairwise touching cubes pairwise intersect. Axis-parallel boxes that pairwise intersect have a common point: project to each coordinate axis and use Helly's theorem on the line (pairwise intersecting intervals have a common point).

Hint 2. So all cubes contain a common point P. For each cube $c+[0,1]^n \ni P$ and each coordinate k, at least one of $[P_k, P_k + \varepsilon]$, $[P_k - \varepsilon, P_k]$ lies in $[c_k, c_k + 1]$ (for small ε). Hence each cube contains one of the 2^n small "orthant boxes" at P.

Hint 3. These orthant boxes have non-empty interior, and the cubes have disjoint interiors, so different cubes contain different orthant boxes: at most 2^n cubes. The 2^n cubes with a vertex at the origin show sharpness.

Problem 3

Hint 1. You must show: for every f there is a convergent sequence of pairwise distinct a_i with $(f(a_i))$ convergent. Look for an infinite set on which both x and $f(x)$ are bounded.

Hint 2. $\mathbb{R} = \bigcup_{m \geqslant 1} \{x \in [-m, m] : \ |f(x)| \leqslant m\}$ is a countable union, and $\mathbb{R}$ is uncountable, so one of these sets is infinite.

Hint 3. Take pairwise distinct points in it; by Bolzano–Weierstrass extract a convergent subsequence, and then a further subsequence along which the bounded values $f(a_i)$ converge.

Problem 4

Hint 1. Encode a figure F by the function $\mathbf{1}_F(x, y)$ and use the rank: a rectangle $I \times J$ gives $\mathbf{1}_I(x)\mathbf{1}_J(y)$, a function of rank one. If F is cut into m rectangles, then $\mathbf{1}_F$ (up to boundaries, e.g. as an integral operator on L^2) has rank at most m.

Hint 2. With the squares $I_i \times J_i$, $\mathbf{1}_F = \sum_{i=1}^n \mathbf{1}_{I_i} \otimes \mathbf{1}_{J_i}$, and this has rank exactly n provided the functions $\mathbf{1}_{I_1}, \ldots, \mathbf{1}_{I_n}$ are linearly independent, and likewise the $\mathbf{1}_{J_i}$. So it suffices to show that intervals of lengths $1, \frac{11}{10}, \ldots, (\frac{11}{10})^{n-1}$ have independent indicators.

Hint 3. If $\sum c_i \mathbf{1}_{I_i} = 0$ with some $c_i \neq 0$, view the intervals with $c_i \neq 0$ as edges between their endpoints; the relation says the c_i form a circulation, so these edges contain a cycle. Going around the cycle, the signed lengths add up to 0: $\sum \pm(\frac{11}{10})^k = 0$ over distinct exponents k.

Hint 4. This is impossible by an 11-adic valuation: the terms have pairwise distinct 11-adic valuations k, so the term with the smallest k cannot be cancelled (multiply by $10^{n-1}11^{-k_{\min}}$ and reduce modulo 11). Hence the rank is n and at least n rectangles are needed.

Problem 5

Hint 1. Let $D = b^2 - 4ac$ (not a square, so $a \neq 0$ and the form vanishes only at $(0,0)$). Take an odd prime $p \nmid aD$ with $\left(\frac{D}{p}\right) = -1$. Show that if $p \mid Q(x,y)$ then $p \mid x$ and $p \mid y$; use $4aQ = (2ax+by)^2 - Dy^2$.

Hint 2. Then $Q(x,y) = p^2 Q(x/p, y/p)$, and by descent $v_p(Q(x,y))$ is even for $(x,y) \neq (0,0)$; hence $v_p(Q(x,y)^z) = z\, v_p(Q(x,y))$ is even too. So a number N with $v_p(N) = 1$ is not representable.

Hint 3. There are infinitely many primes with $\left(\frac{D}{p}\right) = -1$ (quadratic reciprocity plus Dirichlet's theorem, or a direct argument). Choose n of them, $p_1, \ldots, p_n$, not dividing $2aD$, and by the Chinese remainder theorem find N with $N + i \equiv p_i \pmod{p_i^2}$ for $i = 1, \ldots, n$. Then $v_{p_i}(N+i) = 1$ and none of $N+1, \ldots, N+n$ is representable.

Problem 6

Hint 1. Look at the area $s(t)$ of the horizontal cross-section at height t. For the parallelepiped it is constant on $(0, H)$. What is it for a tetrahedron with a horizontal face?

Hint 2. If the tetrahedron has a horizontal face of area A at height h_0 and apex at height h_1, its section at height t between them is a similar triangle of area $A\left(\frac{t-h_1}{h_0-h_1}\right)^2$: a quadratic in t with positive leading coefficient, and 0 outside $[h_0, h_1]$.

Hint 3. On an interval between consecutive heights of vertices, the constant $s(t)$ would equal a sum of such quadratics, at least one of which is present (the section is non-empty). The coefficient of t^2 of the sum is positive: contradiction. (Equivalently, $s'' = \sum s''_T > 0$ there.)

Problem 7

Hint 1. The answer is yes. Look among commuting nilpotent matrices of rank one, e.g. $A = E_{12}$ and $B = E_{13}$ (matrix units); check $AB = BA = 0$.

Hint 2. Both have $A^2 = B^2 = 0$ and rank 1, so in a Jordan basis f_1, f_2, f_3 each is one 2×2 block on two consecutive basis vectors plus a zero block: A sends one basis vector f_{j+1} to f_j and kills the other two, and $\operatorname{im} A = \langle f_j \rangle$.

Hint 3. Here $\operatorname{im} A = \operatorname{im} B = \langle e_1 \rangle$, so in a common Jordan basis the blocks of A and B sit at the same place, and then A and B have the same matrix, i.e. $A = B$. But $A \neq B$.

Problem 8

Hint 1. The image of a continuous map from $[0,1]^2$ is compact and connected, so for an injection it is a non-degenerate segment $[a,b]$. Remove a point.

Hint 2. Choose P with $f(P) \in (a,b)$. The square minus P is connected, but its image $[a,b] \setminus \{f(P)\}$ is not: contradiction.

Hint 3. A continuous bijection $[0,1] \to [0,1]^2$ would be a homeomorphism (compact to Hausdorff); removing an interior point disconnects $[0,1]$ but not the square. A continuous surjection exists (Peano curve), since surjectivity does not see this. The invariant in both cases is connectedness of the complement of a point (cut points).

Problem 9

Hint 1. $\sqrt[3]{2}$ is a root of $x^3 - 2$; $\cos 20^\circ$ satisfies $4c^3 - 3c = \cos 60^\circ = \frac{1}{2}$, i.e. $x = 2c$ is a root of $x^3 - 3x - 1$. A cubic over $\mathbb{Q}$ is irreducible iff it has no rational root.

Hint 2. By the rational root test the candidates are $\pm 1, \pm 2$ and ± 1 respectively, and none works; so both numbers have degree 3.

Hint 3. Theorem: if α is constructible from the points $0, 1$, then $\mathbb{Q}(\alpha)$ lies at the top of a tower $\mathbb{Q} = K_0 \subset K_1 \subset \cdots \subset K_m$ with $[K_{i+1} : K_i] = 2$, so $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is a power of 2 by the tower law. Doubling the cube needs $\sqrt[3]{2}$, trisecting 60° needs $\cos 20^\circ$; degree 3 is not a power of 2.

Problem 10

Note. The quoted argument is in fact a correct proof by contradiction once two implicit steps are justified, so "the error" to be found is the omission of these justifications.

Hint 1. Check each step under the hypothesis $\sum 1/n = S < \infty$. Is regrouping a convergent series allowed? Does a strict termwise inequality between two convergent series give a strict inequality of the sums?

Hint 2. Both are true: regrouping a convergent series does not change its sum, and $\sum(a_k - b_k) > 0$ if all $a_k - b_k > 0$. So the argument is valid; what is missing is exactly these justifications, which is where convergence is used (the same manipulations without the hypothesis would be meaningless).

Hint 3. A direct proof: $s_{2^m} \geqslant 1 + \frac{m}{2}$ by grouping $\frac{1}{2^{j-1}+1} + \cdots + \frac{1}{2^j} \geqslant \frac{1}{2}$, so the partial sums are unbounded. It also gives the growth rate, which is why it is the one to submit.